

A Frequency Dial for Structure-Conditioned Diffusion

SUBMISSION ID

Abstract

A structure-conditioned diffusion adapter is trained for one operating point. Canny edges, a depth map, a sketch: each fixes how tightly the output must follow the condition, and moving that trade-off means training again or scaling the adapter's strength at inference. We train a single adapter on high-pass-filtered image conditions and draw the cutoff r per sample, so the information in the condition varies during training and r becomes a dial at test time. Trained this way, how tightly an output follows its structure is set by the condition while the weights stay fixed. On FFHQ-512, a single adapter moves structure consistency from 0.7469 at $r=0.05$ to 0.2347 at $r=0.3$, each output scored at its own cutoff, and from 0.6524 to 0.2599 with the instrument fixed at $r_m=0.1$, while FID stays between 51.58 and 54.68 over $r \in [0.05, 0.3]$. Scaling this adapter's strength instead pays for lower structure consistency with FID: read with the same fixed instrument, scale 0.50 reaches 0.2146, a little below the cutoff $r=0.3$, and gains diversity, but raises FID to 76.46 against 52.77. On hand drawings filtered at r , the output's correlation with the drawing also falls as r grows.

CCS Concepts

• **Computing methodologies** → **Image manipulation**; **Neural networks**;

1. Introduction

Image editing has long given users a scale or frequency band to turn [FFLS08, PHK11], and sketch-based face synthesis has treated a drawing as a soft constraint rather than a template [CSG*20], in some systems with an explicit control over how far the sketch is trusted [YWL20]. ControlNet [ZRA23] and its successors [MWX*23, ZCC*23, QZY*23, LYK*24] moved that control into training. An adapter trained on Canny edges follows Canny edges; one trained on depth follows depth. Following a structure more loosely then means picking a different checkpoint or scaling the adapter down at inference.

We change what the condition is rather than how hard the model listens to it. Multi-scale editing removes or amplifies a band of the output image [FFLS08, PHK11]; here the band is removed from the condition, and the generative prior decides what fills it. The condition is the image after a radial high-pass filter with normalised cutoff r . At $r=0$ the model receives the grayscale image and the task is colourisation. As r grows the filter removes more of the spectrum, until at $r=1$ almost nothing is left: measured on 500 FFHQ test images, a cutoff of 0.05 already discards 97.5 % of the spectral energy. Training draws r per sample, so one set of weights sees the whole range, and r becomes an inference-time parameter. Trained this way, how tightly an output follows its structure is set by the condition while the weights stay fixed, and scaling the adapter's strength turns out to be a different control (Sec. 4.3).

Our contributions are:

- A control dial. One adapter spans structure consistency 0.7469

to 0.2347 on $r \in [0.05, 0.3]$ with FID between 51.58 and 54.68 (Sec. 4.2).

- The dial and adapter strength separate in FID. With both read by the instrument fixed at $r_m=0.1$, scale 0.50 lowers structure consistency to 0.2146 with more diversity but at FID 76.46, where the cutoff $r=0.3$ reaches 0.2599 at FID 52.77 (Secs. 4.3 and 4.1).
- Measurements behind two design choices (Sec. 5). An encode-decode round trip through the frozen VAE keeps a correlation of 0.5773 with a high-pass condition at $r=0.3$ (Sec. 5.1); on SD1.5 the skip path trained alone, with 6,205,856 parameters, reaches structure consistency 0.5765 at $r=0.1$, where an attention-only design with 6,998,637 parameters reaches 0.2158 (Secs. 5.2 and 5.3).

2. Related work

Conditional control. Text-to-image diffusion [HJA20, SME20, RBL*21, NDR*21, SCS*22] became spatially controllable with ControlNet [ZRA23], which reads the condition image with a small convolutional network, duplicates the encoder and injects residuals through zero convolutions. T2I-Adapter [MWX*23] does the same job with a small feature extractor, IP-Adapter [YZL*23] adds decoupled cross-attention for image prompts, and GLIGEN [LLW*23] grounds generation on boxes. Uni-ControlNet [ZCC*23] and UniControl [QZY*23] fold many condition types into one model, and ControlNet++ [LYK*24] closes a consistency loop on the produced image. A second line shrinks the control branch on the same UNet: ControlNet-XS [ZFR23] couples a much smaller control network to encoder

and decoder in both directions, CtrLoRA [XHSC24] adapts a shared base ControlNet to each new condition with LoRA, and Simple-ControlNet [FCS*25] feeds the condition through a single-scale projection at the top of the UNet with LoRA on self-attention, in under 3M parameters, exposing a scale on that projection at inference. Each of these fixes the condition type at training and, with it, how tightly the output follows; an inference-time scale such as Simple-ControlNet’s moves that operating point by weakening the adapter. Our cutoff moves it through the condition on one set of weights, and on our adapter the two controls separate in FID (Sec. 4.3).

Efficient control on transformer backbones. As backbones moved from UNets to diffusion transformers [PX22, EKB*24], control moved from duplicated encoders to lightweight modules. ControlNeXt [PWZ*24] removes the parallel branch, OminiControl [TLY*24] reuses the backbone itself as the condition processor, and EasyControl [ZYS*25] and NanoControl [LZL*25] inject conditions through LoRA-style [HSW*21] modules at a small fraction of the parameter cost. Recent work also revisits how conditions enter these models: Laytrol [HHLZ25] inherits the backbone’s parameters to preserve pretrained knowledge under layout control, PixelControl [LLZ*26] notes that the VAE’s spatial compression can weaken high-frequency and low-area condition signals, and Zhou et al. [ZWC*26] replace concatenate-and-attend with position-aligned and keyword-scoped attention in multi-condition DiTs. On a UNet the question has a concrete form, since skip tensors are available as an entry point: in our comparison (Sec. 5.2) skip residuals carry more structure than cross-attention.

Frequency bands as an editing axis. Band decompositions are a standard editing axis in graphics: detail manipulation rescales the bands of an edge-preserving or Laplacian decomposition [FFLS08, PHK11], and hybrid images combine the low spatial frequencies of one picture with the high frequencies of another, so that the interpretation changes with viewing distance [OTS06]. In diffusion, frequency decomposition appears mostly as an intervention on the model’s own signals rather than on its condition. FreeU [SHJL23] rebalances skip and backbone features spectrally at no training cost, FreSca [HLT*25] scales frequency bands of the noise prediction using spatial or energy-based cutoffs, DeCo [MWW*25] decouples frequencies in pixel-space diffusion, and FeRA [YHZ*25] routes adaptation by frequency energy. FOCUS [TZY*25] conditions the reverse process on learned frequency priors for test-time adaptation. Closest to us in the choice of control, frequency bands also act as a knob on image-to-image guidance: FCDiffusion [GXZL24] extracts DCT bands of the source features and switches among frequency control branches, FBSDiff [GL24] sets the guiding intensity of a reference through the bandwidth of a substituted band without training, and ILVR [CKJ*21] conditions a pretrained DDPM on a reference at a chosen downsampling factor without further learning. Prasain [Pra26] conditions on wavelet bands instead of edge maps for map-to-satellite synthesis. None of them randomises the condition’s *bandwidth* during training so that the band becomes a control at inference. Randomising a conditioning parameter while training and exposing it at inference is not itself new: SDXL [PEL*23] samples crop coordinates during data loading and surfaces them as a user control, and Ima-

gen [SCS*22] draws the augmentation level of its low-resolution condition at random in training and sweeps it at inference, there to find the best sample quality rather than to give the user a control. Our delta is the choice of parameter and its measured separation from adapter strength: bandwidth and adapter strength are different controls (Sec. 4.3).

Drawings as soft constraints. Sketch-based face synthesis has treated rough drawings as soft constraints [CSG*20] and, closest to our goal, lets the user set how far a drawing is trusted through a refinement level [YWL20]; diffusion-based stroke and sketch synthesis likewise lets the user decide the level of faithfulness to the input [CCC*23]. With pretrained diffusion models, sketch guidance can be applied at inference through a small latent guidance predictor instead of a dedicated encoder [VACO23], and the noise added before denoising balances faithfulness to the input against realism without training [MHS*22]. An extended difference-of-Gaussians operator turns a photograph into line drawings across a range of styles set by its parameters [WKO12]. Our control differs in what it varies: the information content of the condition, set by a frequency cutoff, on one adapter whose response we measure separately from its strength. Because the condition is a continuum and not a fixed extractor, a single adapter can expose it as a dial. StructDiff [HLL*26] targets structure-preserving, spatially controllable single-image generation, and sketch-conditioned pipelines have long used edge maps as the structural carrier.

Training and evaluation. We train with classifier-free guidance [HS22] dropout and Min-SNR loss weighting [HGL*23], and evaluate with FID [HRU*17] under the clean-FID protocol [PZZ21], LPIPS [ZIE*18] and CLIP score [RKH*21] on FFHQ [KLA18] with BLIP [LLXH22] captions.

3. Method

Condition. Given an image x , we convert to grayscale, take the 2D FFT, zero every frequency inside a disc of radius $r \cdot \min(H, W)/2$, invert, and min–max normalise to $[0, 1]$. The normalisation is deliberate: it removes the absolute scale of the filter response, so the model has to infer structure density from spatial statistics. We hypothesise that this is what lets a hand drawing, which carries no r , be used at test time. Figure 1 shows the mask, spectrum and condition at each cutoff.

Condition encoder. A convolutional encoder reads the condition image directly at 512^2 and reduces it $8\times$ to the latent grid: seven 3×3 convolutions with SiLU activations, three of them with stride 2 and followed by group normalisation, widening from 32 to 128 channels before a final projection to 4. Each attention site reads the encoder output through a 1×1 projection. The ablations call an optional 3×3 convolution of 64, 128 or 256 channels inserted before that projection a *stem*; the main model has none. We read the condition with this convolutional encoder rather than the frozen VAE; Sec. 5.1 reports the round-trip measurement.

Injection. Encoder features enter the UNet in two places (Figure 2). Zero-initialised 1×1 heads produce one residual per decoder skip tensor and one for the mid block, added through the UNet’s

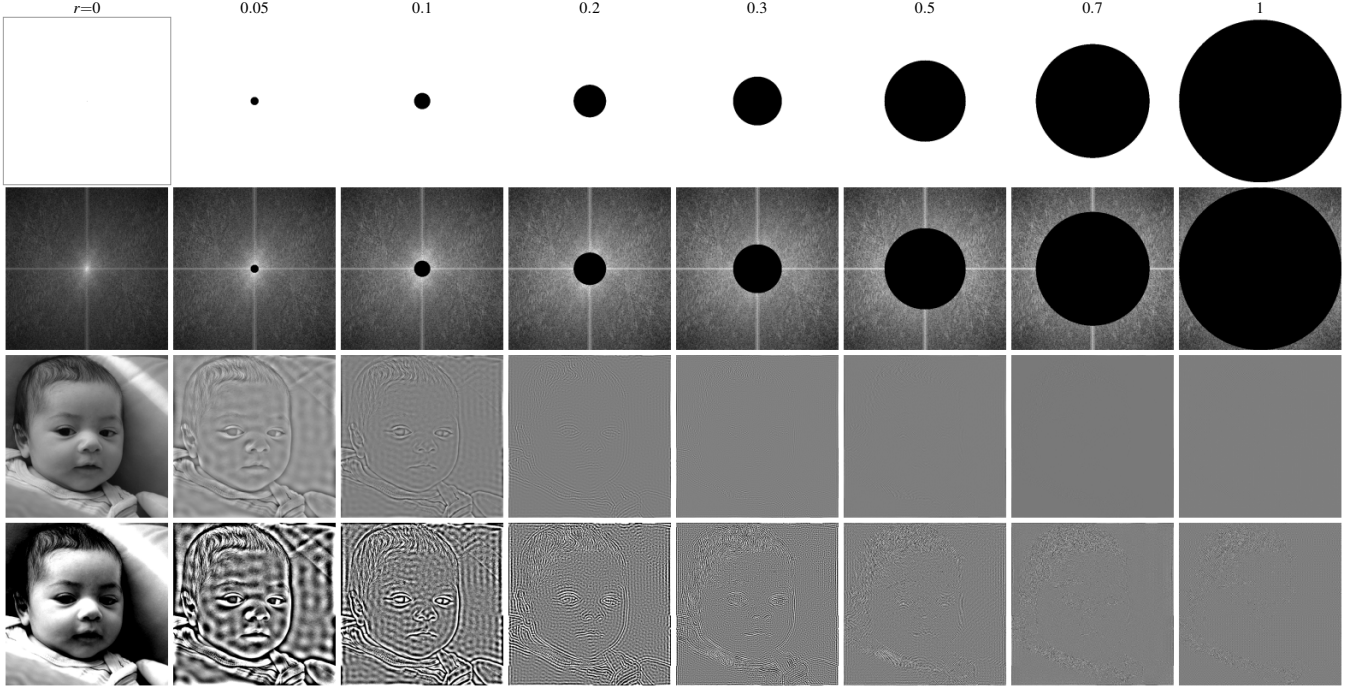


Figure 1: The dial, from the frequency side. Columns are $r = 0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0$. **Row 1:** the radial mask; black is removed. **Row 2:** the log-magnitude spectrum after masking, with the hole growing outward. **Row 3:** the resulting condition, exactly as the model receives it. **Row 4:** the same data displayed at $\pm 2\sigma$ so that the residual content is visible. Above $r \approx 0.2$ the condition looks like flat grey (averaged over 500 test images, its standard deviation falls from 0.238 at $r=0$ to 0.034 at $r=1$), yet the face outline survives in row 4, and the model still reaches SC 0.4019 at $r=0.2$. The per-image normalisation leaves the model only this low-contrast residue to read structure from.

existing ControlNet interface. In parallel, each cross-attention site attends over a 3×3 window of the condition features, gated by a zero-initialised per-channel vector. At 1×1 the softmax runs over a single position, the site reduces to a gated value projection, and structure consistency is no lower (Sec. 5.2). Both paths start as exact no-ops: with them switched off the network reproduces stock Stable Diffusion 1.5 [RBL*21] bit for bit, which we assert before every training run.

Training. SD1.5 [RBL*21] stays frozen. We train 6,611,469 parameters on FFHQ-512 [KLA18] (images under their Flickr authors’ Creative Commons or public-domain licences, the dataset under CC BY-NC-SA 4.0) with BLIP [LLXH22] captions, split 65,100 train / 1,400 validation / 3,500 test by a seeded shuffle of the 70,000 images (5% test, 2% validation, seed 2026). AdamW, learning rate 10^{-4} , $\beta = (0.9, 0.999)$, cosine schedule with 500 warm-up steps, gradient clipping at 1.0, batch 32, bf16, 12 epochs on one H200. The loss is ϵ -prediction with Min-SNR weighting [HGL*23], $\gamma = 5$. Text conditioning is dropped with probability 0.1 and the structure condition independently with probability 0.05. AdamW uses the PyTorch default weight decay of 0.01. An EMA of the adapter (decay 0.9999) is kept with a 10-step interval; evaluation uses the raw weights. r is drawn uniformly from $[0, 1]$ per sample, although we claim the dial only on $r \in [0.05, 0.3]$. Appendix B lists the remaining settings.

4. Experiments

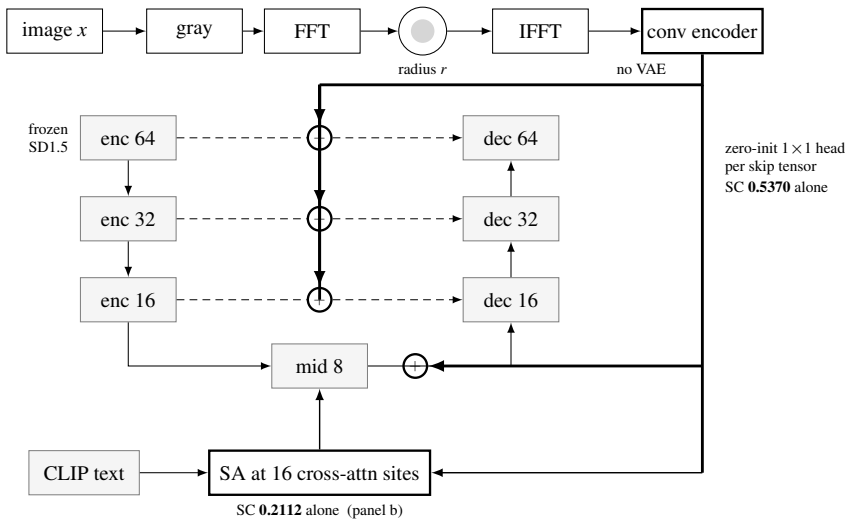
4.1. Metrics

We evaluate on 500 of the 3,500 held-out images with a DDIM sampler at 50 steps, CFG 7.5, seed 2026.

Structure consistency. SC converts the generated image to grayscale, applies the *same* high-pass filter at the *same* cutoff the model was given, and takes the Pearson correlation against the given condition over the flattened pixel vector, per image, averaged over images. Both signals are the per-image min–max normalised filter output, so the correlation is scale free. Two anchors bound it: feeding the ground truth back gives exactly 1.000 at every r , and scoring a generation against the condition of a different test image gives 0.0847 at $r=0$ and 0.0082 at $r=0.1$. Because the instrument moves with r , a row-to-row difference mixes model behaviour with instrument change, so we report a cutoff-independent metric beside it and a fixed-instrument reading below.

Edge-F1. Canny edges (thresholds 100 and 200 on the 8-bit grayscale image) are taken from the generation and from the reference and matched with a 2-pixel tolerance, implemented as a 5×5 dilation of each map before intersecting; F1 is computed per image and averaged. It does not depend on r , which is why it appears

(a) condition path and the two injection sites



(b) inside one site

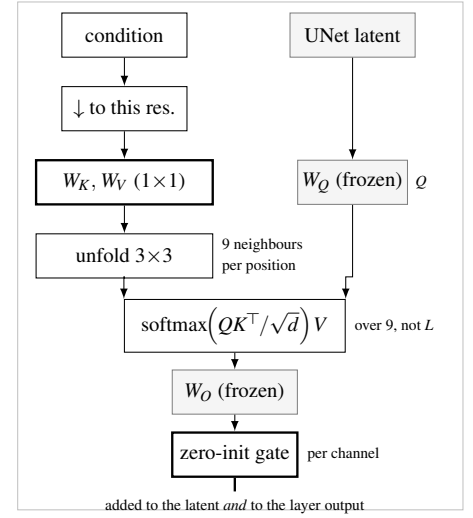


Figure 2: (a) A disc of radius r is masked out of the spectrum and a trainable convolutional encoder reads the result at full resolution, never through the VAE. Its features enter the frozen UNet twice: as zero-initialised residuals on every skip tensor ($\oplus$), and through StandAlone attention at the 16 cross-attention sites. Stroke weight follows the measured split: at $r=0$, SC is 0.5370 with only the skip path active and 0.2112 with only attention (Sec. 5.2). (b) One StandAlone site. The query comes from the UNet’s own latent through the frozen W_Q ; keys and values come from the condition through trained 1×1 projections, unfolded so each position attends over its 3×3 neighbourhood rather than the whole map; a 1×1 window performs at least as well (Sec. 5.2). The result passes the frozen output projection and a zero-initialised per-channel gate, then is added both into the latent — so the text query below sees structure — and to what the layer returns.

in Table 1: 0.7796 at $r=0.05$ falling to 0.4724 at $r=1$, so the dial survives a cutoff-independent instrument.

Diversity. Mean pairwise LPIPS (AlexNet backbone) over $K=4$ samples generated from one condition with different seeds; higher means the condition constrains less. It is not independent of the LPIPS column (defined below): across the sweep LPIPS – Diversity is 0.0320 ± 0.0038 , nearly constant while Diversity spans 0.21, so Diversity carries close to the same information as LPIPS and serves as a secondary reading.

FID, LPIPS and CLIP. Clean-FID [HRU*17,PZZ21] in its clean mode at 512 pixels, between the 500 generations and the same 500 references. The LPIPS column is AlexNet LPIPS between each generation and its own reference, averaged over images. CLIP score is 100 times the cosine similarity of ViT-B/32 [RKH*21] embeddings of the image and its own caption.

Fixed instrument. Structure consistency re-extracts the condition at the same cutoff the model was given, so the measuring instrument moves with the thing it measures. We therefore also re-score the saved generations with the cutoff held at $r_m=0.1$ throughout. This reads the saved 8-bit images, so at $r=0.1$ it gives 0.6210 where Table 1, computed before quantisation, gives 0.6211.

4.2. The dial

Table 1 gives the operating curve. On $r \in [0.05, 0.3]$ structure consistency falls at every step while FID stays between 51.58 and

54.68, and sample diversity rises alongside. Going from $r=0$ to $r=0.05$ structure consistency rises by 0.0479 ($z = 6.24$ with the unpaired standard error over the 500 images), so the sweep is not monotone at its left end; the same rise appears in Edge-F1 (0.7776 to 0.7796), so it is model behaviour rather than an artefact of the SC instrument. And at the right end $r=0.5$ and $r=0.7$ are not separable across seeds (paired $t = 0.732$, $df = 2$, critical 4.303). FID stays between 50.13 and 57.40 across the entire sweep.

Figure 3 holds the seed and the caption fixed and varies only r . Because the caption is derived from the reference, it could be carrying part of the structure. We repeated the sweep with one generic caption, “a photo of a person”, for every image. Structure consistency stays close (0.7039, 0.6179 and 0.2196 at $r=0$, 0.1 and 0.3, against 0.6991, 0.6211 and 0.2347 with captions) while FID rises to 55.56–65.84, as it should once the caption stops describing the image. The structure comes from the condition. Figure 4 plots the curve and, below it, the cutoff and the two scaled adapters in the structure-consistency/FID plane.

Across seeds 2026, 2027 and 2028 the curve is stable: SC at $r=0.05$ is 0.7479 ± 0.0015 and at $r=0.3$ it is 0.2318 ± 0.0106 .

Re-scored with the fixed instrument of Sec. 4.1, the curve still falls: 0.6524 at $r=0.05$, 0.6210 at $r=0.1$ and 0.2599 at $r=0.3$. Holding the instrument fixed compresses the range, as it must, and keeps the decline, so the decline comes from the generations rather than from the metric.

beyond the claimed range

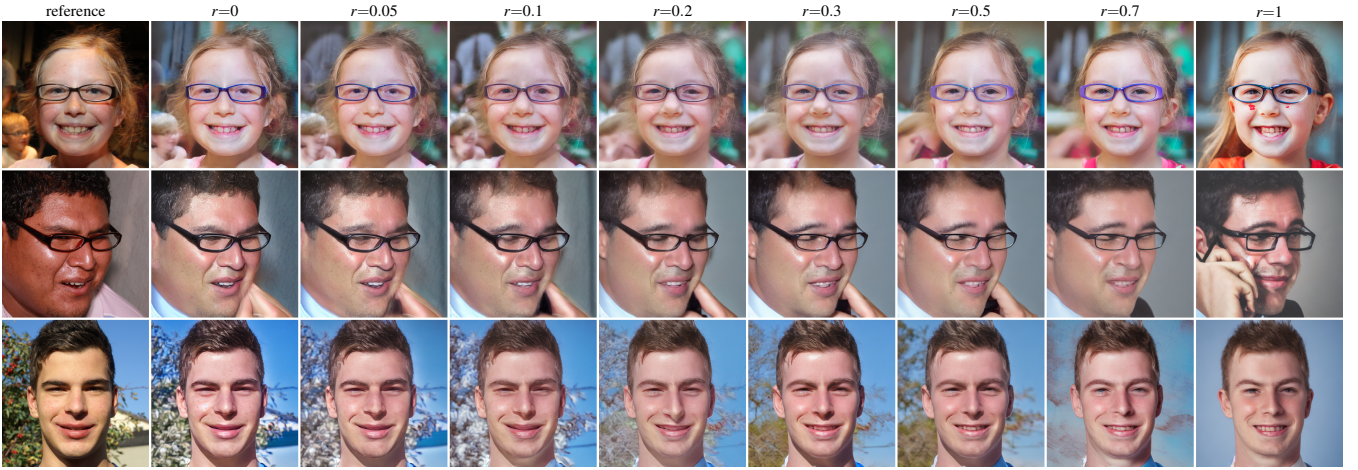


Figure 3: One seed and one caption per row; only r changes. The rows are the first three test images in generation order. As r rises, background detail thins and clothing and colour drift from the reference while the face layout holds; at $r=1$ the pose of the second row changes as well. The reference is never shown to the model. Columns beyond $r=0.3$ lie outside the claimed range; at $r=1$ colour artefacts appear on the first row.

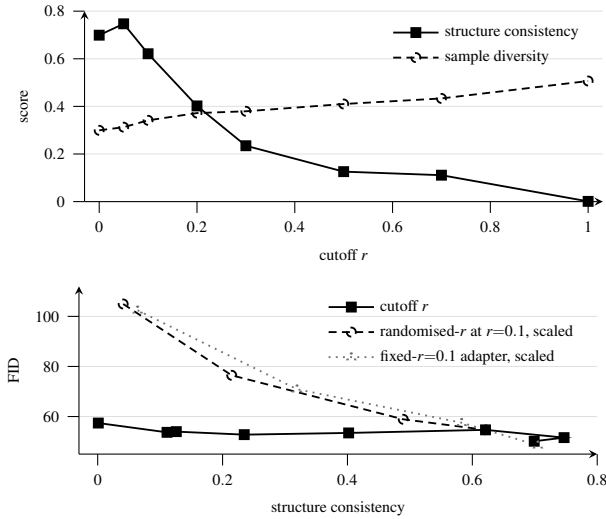


Figure 4: *Top:* structure consistency falls and sample diversity rises as the cutoff grows. *Bottom:* the cutoff sweep of Table 1 and the two scaled adapters of Table 2 in the structure-consistency / FID plane. Narrowing the condition’s bandwidth lowers structure consistency within a narrow FID band; weakening either adapter lowers it while FID rises. Scaled points are scored at $r_m=0.1$, cutoff points at their own cutoff. Both panels are plotted directly from the evaluation outputs.

4.3. Adapter strength as the alternative knob

An obvious alternative is to train on one cutoff and turn the adapter’s contribution down at inference, the knob lightweight control modules expose [FCS*25]. We ran it on the adapter trained at

r	SC $\pm$ SEM	seed sd	Edge-F1	Diversity	FID
0.00	0.6991 ± 0.0070	0.0105	0.7776	0.2994	50.13
0.05	0.7469 ± 0.0030	0.0015	0.7796	0.3126	51.58
0.10	0.6211 ± 0.0035	0.0059	0.7439	0.3414	54.68
0.20	0.4019 ± 0.0037	0.0098	0.7113	0.3719	53.45
0.30	0.2347 ± 0.0041	0.0106	0.6878	0.3794	52.77
<i>beyond the range we claim</i>					
0.50	0.1259 ± 0.0047	0.0093	0.6225	0.4100	53.97
0.70	0.1110 ± 0.0047	0.0186	0.5788	0.4334	53.69
1.00	0.0010 ± 0.0006	0.0002	0.4724	0.5066	57.40

Table 1: Operating curve, 500 of the 3,500 test images. SEM is over those 500 images; seed sd is over three training seeds. Edge-F1 is a cutoff-independent reading of the same runs and is reported alongside SC because SC re-extracts at the same cutoff the model was given (Sec. 4.1). FID has a single estimate per cell and no variance, so small differences between rows are not read as effects.

fixed $r=0.1$ (Sec. 4.4), scaling both injection paths, and, as a second check, scaled the randomised- r model at the same cutoff (Table 2; Table 10 sweeps it finely).

Both scaled curves leave the FFHQ image distribution as they go. At scale 0.25 the fixed- r adapter reaches FID 102.43 and diversity 0.7689, beyond anything the cutoff produces over its whole sweep, where FID stays within 50.13–57.40 and diversity peaks at 0.5066. Turning the adapter down removes structure while FID rises, to 76.46 at scale 0.50; narrowing the condition’s band removes it while FID stays between 51.58 and 54.68 on the claimed range. At full strength the fixed- $r=0.1$ adapter reaches FID 48.51 against 54.68 for the randomised model at the same cutoff. Figure 5 shows what the scaled settings produce for the first two test images.

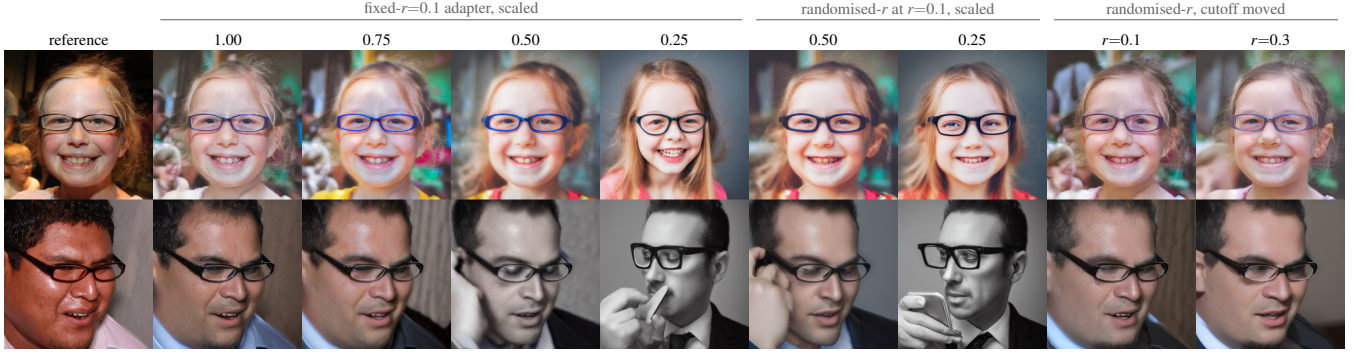


Figure 5: The first two test images under adapter strength and under the cutoff, one seed and caption per row, taken from the runs of Table 2. Lowering the strength of either adapter yields photographs whose pose, framing and colour depart from the reference (the second row turns grayscale at scale 0.25), while moving the cutoff to $r=0.3$ keeps the reference layout.

Setting	SC	Diversity	FID
<i>fixed-r=0.1 adapter, scaled</i>			
scale 0.25	0.0640	0.7689	102.43
scale 0.50	0.3193	0.5540	70.91
scale 0.75	0.5829	0.4044	57.52
scale 1.00	0.7050	0.2967	48.51
<i>randomised-r adapter at $r=0.1$, scaled</i>			
scale 0.25	0.0408	0.7752	104.95
scale 0.50	0.2146	0.6061	76.46
scale 0.75	0.4893	0.4171	58.90
<i>randomised-r adapter, cutoff moved</i>			
$r=0.2$	0.4019	0.3719	53.45
$r=0.3$	0.2347	0.3794	52.77
<i>same generations, instrument fixed at $r_m=0.1$</i>			
$r=0.05$	0.6524	0.3126	51.58
$r=0.2$	0.3838	0.3719	53.45
$r=0.3$	0.2599	0.3794	52.77

Table 2: Strength against cutoff. Every scaled row is generated and scored at $r=0.1$, so it already uses the instrument $r_m=0.1$; the cutoff-moved rows are scored at their own cutoff, and the last block re-scores the same generations with $r_m=0.1$ (Sec. 4.1). With that one instrument, scale 0.50 of the randomised- r adapter reaches 0.2146 at FID 76.46, and the cutoff $r=0.3$ reaches 0.2599 at FID 52.77. Scale 1.00 of the fixed- r adapter is a separate inference run of the specialist of Sec. 4.4 (0.7049 there).

4.4. Against per-cutoff specialists

We trained four adapters at fixed r and compared each against the single randomised- r model at its own cutoff. Because the single model was trained three times, we divide each gap by the single model’s three-seed standard deviation rather than by a within-run standard error, and as a conservative rule treat a ratio below the two-sided t critical value at $df = 2$ (4.303) as not separable. The result is mixed: the specialist wins at $r=0.1$ (0.7049 against 0.6245, $\Delta = +0.0803$, $+13.55$ sd) and the single model wins at $r=0.3$ (0.2318

against 0.1214, -0.1104 , -10.42 sd). Neither $r=0.5$ (-1.00 sd) nor $r=0.7$ ($+0.14$ sd) separates (Table 11).

One model does not dominate four. It gives a dial the specialists cannot: a model trained at $r=0.1$ has never seen $r=0.3$, so its curve is a single point.

4.5. Baselines

Against pretrained ControlNet [ZRA23], over the same test images and captions, each baseline is given Canny edges (thresholds 100 and 200) at conditioning scale 1.0, with the pipeline’s default PNDM sampler, 50 steps, CFG 7.5 and seed 2026. ControlNet-Canny reaches Edge-F1 0.6854, LPIPS 0.5294, FID 71.92 with 361,279,120 trainable parameters; the ControlNet-HED checkpoint, fed the same Canny edges rather than HED boundaries, reaches 0.5733 / 0.6548 / 118.98. At $r=0.1$ our adapter reaches Edge-F1 0.7439, LPIPS 0.3748 and FID 54.68 with 6,611,469 parameters. Two asymmetries make this a comparison of operating points rather than a like-for-like control: the conditions differ (Canny edges against a frequency band), and our adapter was trained on 65,100 FFHQ images while ControlNet is zero-shot here and FID is measured against FFHQ statistics. On text alignment ControlNet-Canny’s CLIP 30.34 exceeds our best over the whole sweep, 30.30 (ControlNet-HED reaches 29.16). T2I-Adapter-Canny [MWX*23], a 77,000,640-parameter side network and so closer to our design than a copied encoder, reaches Edge-F1 0.5845, LPIPS 0.5690, FID 86.66 and CLIP 30.58 under the same protocol. We leave out its sketch checkpoint, which expects PiDiNet sketches as input; fed Canny edges it would measure that mismatch rather than the method.

4.6. Outside the training domain

Everything above is FFHQ. To test whether the dial is specific to faces we ran the sweep on the 500 general-scene photographs of the public photograph-sketch set rhfeiyang/photo-sketch-pair-500 on Hugging Face, centre-cropped to 512², and, as an in-domain reference under the identical script, on the 500 FFHQ test images, both with the prompt “a high-quality photo”. Each



Figure 6: Hand-drawn face sketches (top row) and what the model makes of them (bottom row), zero-shot. The drawings carry no cut-off; the model has to read structure density off the drawing itself, which the per-image min-max normalisation permits. Most of the scans carry the archive filename stamped across the top, and the model redraws it — a literal reading of the condition it was given. Removing the whole top 20% of the frame (102 rows), a generous over-estimate of that band, moves the score to 0.5398 (-0.0086, 0.32 standard errors), so the band is not what the score is measuring.

generation is scored in two ways: the grayscale output against the grayscale input photograph, and the grayscale output against the filtered condition it was given. Neither filters the output, so neither is the structure-consistency instrument of Table 1, and the in-domain values differ from it (Table 12).

Against the input photograph the score falls from 0.4506 at $r=0.05$ to 0.1384 at $r=0.3$ out of domain, and from 0.5981 to 0.4142 in domain; against the filtered condition it falls from 0.3139 to 0.0303 out of domain and from 0.3193 to 0.0334 in domain. The dial carries out of domain. With the adapter off, the out-of-domain images score 0.0145 against the input photograph at $r=0.1$, against 0.3106 with it on, so the residual structure comes from the condition and not from the backbone.

Image quality does not transfer: FID, each run measured against its own input photographs, rises from 65.71 in domain to 148.97 out of it at $r=0.1$. The control survives the domain shift; the generative quality does not.

4.7. Hand-drawn sketches

Figure 6 shows the transfer. On 40 real face drawings from the CUHK Face Sketch database [WT09], which the model was never trained on, prompted with “a high-quality photo of a face”, the grayscale output correlates with the input drawing at 0.5484 and CLIP score is 23.43. FID is not reported: 40 images is too few for a stable estimate, and our evaluator refuses rather than print one. We also ran 500 general-scene drawings, the sketch half of the public photograph-sketch set `rhfeiyang/photo-sketch-pair-500` on Hugging Face, far outside the training domain and with the same face prompt, where that correlation falls to 0.1714 and FID against FFHQ is 285.91: the model follows the drawing but does not produce a face. It was trained on nothing else.

A drawing arrives without a cutoff, but the artist can give it one:

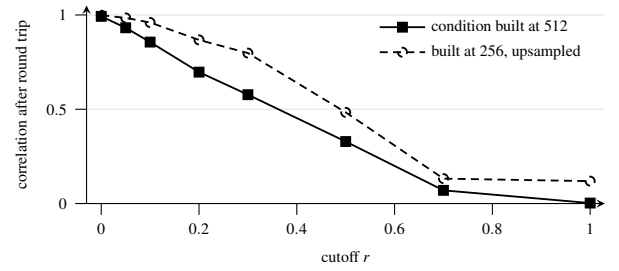


Figure 7: Correlation between the condition fed to the autoencoder and the same condition re-extracted after an encode-decode round trip, 24 images. Above $r \approx 0.5$ the condition is effectively gone. Building the condition at 256 and upsampling puts the structure at lower absolute frequency and preserves far more, which is why comparisons against older pipelines need the source resolution stated.

filter the drawing at r and hand the model the result, the operation every training condition went through. On 40 face drawings, the grayscale output’s correlation with the original drawing falls from 0.5486 at $r=0$ to 0.3390 at $r=0.1$, 0.2271 at $r=0.3$ and 0.0407 at $r=1$; $r=0.5$ and $r=0.7$ lie within a standard error (0.0419) of each other.

Against the filtered drawing the model was actually given, the same generations score 0.2552 at $r=0.05$, 0.1399 at $r=0.1$ and 0.0142 at $r=0.3$. Scored by the same script, FFHQ test photographs reach 0.3193, 0.1796 and 0.0334 (Sec. 4.6), so a filtered drawing is followed somewhat less closely than a filtered photograph and declines the same way (Table 13).

5. What the design needs

5.1. How much of the condition survives the frozen VAE

Figure 7 measures how much of a high-pass condition survives the frozen SD VAE. After an encode-decode round trip, the re-extracted condition correlates with the original at 0.8562 for $r=0.1$, 0.5773 for $r=0.3$ and 0.0700 for $r=0.7$. The round trip lower-bounds what the latent retains: a component may survive in the latent that the frozen decoder cannot render. The frozen-VAE designs in Table 3 inject through attention only and reach structure consistency 0.0177 at $r=0.1$, where the round trip still keeps a correlation of 0.8562, so condition loss in the round trip does not by itself account for their result. An image-prompt design such as IP-Adapter [YZL*23] reads a CLIP image embedding instead, so this measurement does not speak to it. Building the condition at 256 and upsampling moves the structure to lower absolute frequency, and the round trip then keeps 0.7978 at $r=0.3$ against 0.5773 at native resolution, a data-pipeline artefact worth knowing about when comparing against older setups.

5.2. Where to inject

Table 3 lists absolute configurations; each row below the control differs from the *conv* / *attn* row in one field. Figure 8 decomposes the main model by switching paths off at test time. Keeping only

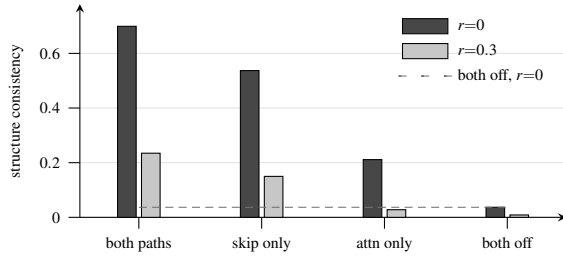


Figure 8: Both injection paths on, then each alone, then neither. The skip residuals carry most of the structure signal; cross-attention alone recovers little of it, and with both off the model falls to the frozen backbone (FID 120.50 against 50.13 with both paths on; skip only 65.12, attention only 67.07). The dashed line is the both-off control at $r=0$. At $r=0$ the single-path gains over the control roughly add up to the both-paths score; at $r=0.3$ both paths together exceed that sum.

cross-attention drops SC at $r=0$ from 0.6991 to 0.2112; keeping only the skip path retains 0.5370. With both paths off the model falls to 0.0368 SC and FID 120.50, the control that says the adapter is doing the work. On FID the two single paths are close (65.12 against 67.07), so the placement effect we measure is a structure-fidelity effect and not a general image-quality one.

Within the attention-only design, a 128-channel stem raises SC at $r=0.05$ from 0.1485 to 0.2826, and a 256-channel stem reaches 0.2986 with 6,998,637 trainable parameters against 405,613 without a stem; the skip path trained alone, with 6,205,856 parameters, reaches 0.7085 (Sec. 5.3). A 7×7 window gives 0.0359, far below the 3×3 design (0.1485) and close to the no-adapter control (0.0074). All of these vary capacity with the skip path absent, so they measure capacity within the attention-only design.

Table 4 repeats the one-field ablations on the main model with both paths on. Each is a single seed, so we read it directionally against the main model’s three-seed spread at each cutoff (0.0105 at $r=0$, 0.0015 at $r=0.05$). Capacity helps here too. At $r=0$ a 256-channel stem lifts SC from 0.6991 to 0.7900, several times the seed spread there; adding the skip path to attention alone moves it from 0.2112 to 0.6991. At $r=0.05$ the stem gain shrinks (0.7469 to 0.7631), and FID at $r=0.1$ drops from 54.68 to 51.36. The attention window does not help: 1×1 matches 3×3 within the seed spread or exceeds it at every cutoff listed, and 5×5 is lower at all of them, so what the attention sites contribute is where the condition enters, not a neighbourhood. The zero-initialised gate is worth keeping; without it SC is lower from $r=0$ to 0.3 and FID at $r=0.1$ is higher. The largest single change is how r is drawn. A log-uniform sampler lifts SC at $r=0$ to 0.9480 and lowers FID there to 42.06, but gives up the upper range (0.0858 at $r=0.5$ against 0.1259): it moves the dial’s resolution toward small r rather than improving it everywhere. The results above use the uniform sampler. The main model keeps the 3×3 window and no stem, although the 1×1 and stem variants score higher at several cutoffs in Table 4, because these one-field ablations were trained after the main configuration and its three seeds had been fixed.

Cond. encoder	Injection	SC $r=0.05$	SC $r=0.1$	FID $r=0.1$
no adapter (control)		0.0074	0.0082	120.50
frozen VAE	attn, no zero-init gate	0.0234	0.0147	76.25
frozen VAE	attn	0.0303	0.0177	65.71
conv	attn	0.1485	0.0772	56.39
conv + 128-ch stem	attn	0.2826	0.1882	53.17
conv + 256-ch stem	attn	0.2986	0.2158	52.84
conv	attn, 7×7	0.0359	0.0218	58.51
conv	attn + skip	0.7469	0.6211	54.68

Table 3: Absolute configurations, not a cumulative chain: every row below the control differs from the conv / attn row in exactly one field, and all rows use a 3×3 window and a zero-initialised gate unless stated. The main model is the last row and carries no stem; its attention sites read the encoder output through a 1×1 projection. The frozen-VAE rows use attention injection only. They and the 7×7 row stay close to the no-adapter control at both cutoffs, so at that level a weak design and a run that failed to learn are not distinguishable. Stem width with the skip path present is varied in Table 4.

One field changed	SC				FID
	$r=0$	0.05	0.3	0.5	$r=0.1$
main model	0.6991	0.7469	0.2347	0.1259	54.68
seed sd (3 seeds)	0.0105	0.0015	0.0106	0.0093	–
1×1 window	0.7825	0.7640	0.2271	0.1255	55.70
5×5 window	0.6764	0.7175	0.2039	0.0944	55.68
no zero-init gate	0.6838	0.7322	0.2058	0.1290	57.02
average-pool downsampling	0.7203	0.7465	0.2279	0.1241	55.13
64-ch stem	0.7775	0.7594	0.2354	0.1170	51.91
256-ch stem	0.7900	0.7631	0.2511	0.1214	51.36
colour condition	0.6949	0.7463	0.2343	0.1255	54.75
no condition dropout	0.7042	0.7353	0.2279	0.1231	54.69
$r=u^2$ sampler	0.8936	0.7836	0.2468	0.1039	53.46
log-uniform sampler	0.9480	0.8000	0.2302	0.0858	54.87

Table 4: One-field ablations on the main model, skip and attention paths both on, one training seed each; the second row is the main model’s spread over three seeds. The main model uses a 3×3 window, a zero-initialised gate, learned 2×2 strided downsampling of the condition features at each attention site, no stem before the 1×1 projection, a grayscale condition, condition dropout 0.05, and r uniform on $[0, 1]$. Drawing $r = u^2$ with u uniform concentrates training at small r ; the log-uniform sampler on $[10^{-3}, 1]$ does so more strongly.

5.3. The skip path alone

Section 5.2 ablates the two injection paths on a model trained with both. That confounds placement with capacity: the cross-attention branch is still present and trained, just switched off at test time. We therefore trained the skip path from scratch, with the attention branch removed entirely.

It reaches SC 0.5765 at $r=0.1$ against 0.6211 for both paths and 0.0275 for cross-attention alone, at FID 58.20 against 54.68. Trained on its own, the skip path recovers most but not all of the

r	skip only	both paths	attention only
0.05	0.7085	0.7469	—
0.10	0.5765	0.6211	0.0275
0.20	0.3508	0.4019	—
0.30	0.1608	0.2347	0.0279
0.50	0.0709	0.1259	—
0.70	0.0757	0.1110	—

Table 5: Structure consistency for a model trained with the skip path only, against the main model and against its cross-attention branch measured in isolation. The last column comes from switching a path off at test time, so it is not a trained-from-scratch comparison.

combined model, so the attention branch adds something the skip path does not supply. The skip-only model trains 6,205,856 parameters against 6,611,469 for both paths, and the attention-only design with a 256-channel stem, at 6,998,637 parameters, reaches 0.2158 at $r=0.1$ (Table 3). The full curve is in Table 5.

6. Limitations

Everything here is one domain (FFHQ faces) and one backbone (SD1.5), and the three results depend on that backbone to different degrees. The dial is defined on the condition alone; we measured it on SD1.5 with FFHQ-512. SDXL [PEL*23] keeps the UNet skip tensors and is the closest next test. The skip-path injection is specific to UNets: MMDiT backbones [EKB*24] have no UNet skip connections, and control there enters through shared tokens or LoRA [HSW*21] modules [TLY*24,ZYS*25,LZL*25]. The VAE measurement covers the four-channel SD autoencoder.

We claim the dial on $r \in [0.05, 0.3]$. Above that range the condition retains little energy (0.0016 of the spectrum at $r=0.5$), and structure consistency becomes hard to interpret, since two nearly empty signals correlate near zero whatever the model does. FID is a single estimate per cell with no variance, so we leave gaps inside the band of the cutoff sweep (50.13–57.40) uninterpreted. The strength setting’s FID is measured against FFHQ statistics; the samples of Figure 5 depart from the reference layout rather than visibly degrading, so the rise reflects leaving that distribution at least in part, and we report no no-reference quality measure. The one-field ablations of Table 4 are single-seed, and the per-image normalisation of the condition was fixed by design and never ablated. At $r=0.1$ the fixed-cutoff specialist leads the single model by +13.55 seed sd (Sec. 4.4). The frozen VAE was never paired with the skip path, so in Table 3 the VAE result is confounded with the weakness of attention-only injection. The sketch results have no adapter-off control. The ControlNet comparison is between operating points; a ControlNet trained on our condition and data would be the like-for-like control, and we did not train one. We do not compare against a sketch-reliability control [YWL*20] or against noise-level editing [MHS*22]; both act on different inputs and would need their own protocol.

The adapter trains 6,611,469 parameters, fewer than ControlNet and more than recent lightweight control modules: Simple-ControlNet [FCS*25] reports under 3M parameters on SD1.5,

ControlNet-XS [ZFR23] ships much smaller control networks than ControlNet, and the DiT modules above use a fraction of our count. We make no efficiency claim. We evaluate the dial with image metrics only; whether equal steps in r are perceived as equal changes, and whether artists find the control useful, is not measured here.

7. Conclusion

Randomising the bandwidth of a structure condition during training turns the cutoff into an inference-time control, and that control is not reproducible by scaling this adapter’s strength. A single adapter, trained once with the cutoff drawn per sample, moves structure consistency from 0.7469 to 0.2347 as r goes from 0.05 to 0.3, each output scored at its own cutoff (Table 1). Read with the instrument fixed at $r_m=0.1$, scaling its strength to 0.50 reaches 0.2146 at FID 76.46, where the cutoff $r=0.3$ reaches 0.2599 at FID 52.77 (Sec. 4.3). The design reads the condition with a convolutional encoder rather than the frozen autoencoder and injects it through the UNet skip tensors and cross-attention, both choices measured on SD1.5. One checkpoint covers the cutoff range $r \in [0.05, 0.3]$ at FID between 51.58 and 54.68, where the specialists of Sec. 4.4 need one adapter per operating point.

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variant	$r=0$	0.05	0.1	0.3	0.5
main model	0.6991	0.7469	0.6211	0.2347	0.1259
1×1 window	0.7825	0.7640	0.6382	0.2271	0.1255
5×5 window	0.6764	0.7175	0.5919	0.2039	0.0944
64-ch stem	0.7775	0.7594	0.6358	0.2354	0.1170
256-ch stem	0.7900	0.7631	0.6332	0.2511	0.1214
average-pool downsampling	0.7203	0.7465	0.6181	0.2279	0.1241
colour condition	0.6949	0.7463	0.6251	0.2343	0.1255
log-uniform sampler	0.9480	0.8000	0.6617	0.2302	0.0858
$r=u^2$ sampler	0.8936	0.7836	0.6568	0.2468	0.1039
no condition dropout	0.7042	0.7353	0.6108	0.2279	0.1231
no zero-init gate	0.6838	0.7322	0.6118	0.2058	0.1290
<i>designs not adopted (Sec. 5.1)</i>					
256px source pipeline	—	0.0271	0.0171	0.0291	—
VAE condition	—	0.0234	0.0147	0.0172	—

Table 6: Structure consistency for every one-field ablation, at every measured cutoff. Each row changes one field of the main model and nothing else. The last block holds two designs the paper argues against: building the condition at 256 and upsampling (FID 86.84 at $r=0.1$), and reading it through the frozen VAE.

variant	$r=0$	0.05	0.1	0.3	0.5
main model	50.13	51.58	54.68	52.77	53.97
1×1 window	49.13	52.34	55.70	54.16	54.33
5×5 window	52.05	54.16	55.68	54.43	55.39
64-ch stem	48.92	50.25	51.91	51.08	52.03
256-ch stem	47.74	49.49	51.36	51.77	52.13
average-pool downsampling	50.90	53.43	55.13	54.29	54.32
colour condition	52.53	52.09	54.75	52.67	52.84
log-uniform sampler	42.06	52.17	54.87	54.33	56.96
$r=u^2$ sampler	44.69	51.89	53.46	52.87	54.52
no condition dropout	51.17	52.50	54.69	53.13	54.26
no zero-init gate	52.72	55.63	57.02	54.53	55.51

Table 7: FID for the same one-field ablations as Table 6, one training seed each. FID is a single estimate per cell with no variance (Table 1), so small differences are not read as effects; the largest departures are the log-uniform and $r=u^2$ samplers at $r=0$ (42.06 and 44.69 against 50.13).

Appendix A: Full per-cutoff results

The main text reports each experiment at the cutoff that carries its argument. The tables here give every cutoff that was measured, from the same `results.json` files the rest of the paper is generated from. Tables 6 and 7 extend Table 4; Table 8 extends Table 1; Table 9 extends Sec. 5.3 and the seed spread; Table 10 extends Sec. 4.3; Table 11 extends Sec. 4.4; Tables 12 and 13 extend Sec. 4.6 and the hand-drawn sketches; Table 14 lists the settings behind all of them.

Appendix B: Reproducibility

r	SC	FID	LPIPS	CLIP	Edge-F1	Div.
0.00	0.6991	50.13	0.3341	29.54	0.7776	0.2994
0.05	0.7469	51.58	0.3495	29.41	0.7796	0.3126
0.10	0.6211	54.68	0.3748	29.53	0.7439	0.3414
0.20	0.4019	53.45	0.4003	29.47	0.7113	0.3719
0.30	0.2347	52.77	0.4148	29.73	0.6878	0.3794
0.50	0.1259	53.97	0.4418	29.78	0.6225	0.4100
0.70	0.1110	53.69	0.4625	29.92	0.5788	0.4334
1.00	0.0010	57.40	0.5329	30.30	0.4724	0.5066

Table 8: Every metric for the main model, at every cutoff. The SC and FID columns are the ones plotted in Figure 4.

r	skip only	skip only FID	seed mean	seed s.d.
0.00	0.6284	55.84	0.7109	0.0105
0.05	0.7085	57.65	0.7479	0.0015
0.10	0.5765	58.20	0.6245	0.0059
0.20	0.3508	57.07	0.4036	0.0098
0.30	0.1608	57.21	0.2318	0.0106
0.50	0.0709	57.41	0.1304	0.0093
0.70	0.0757	57.95	0.1261	0.0186
1.00	0.0006	59.60	0.0012	0.0002

Table 9: Left: the skip path trained alone (Sec. 5.3). Right: mean and standard deviation of structure consistency over seeds 2026, 2027 and 2028 for the main model.

adapter strength	SC	FID	SC, seed 2027
0.1	0.0165	117.58	—
0.2	0.0300	110.20	—
0.25	0.0408	104.95	0.0309
0.35	0.0808	95.62	—
0.5	0.2146	76.46	0.2038
0.6	0.3373	66.39	—
0.75	0.4893	58.90	0.4823
1.00	0.6211	54.68	—

Table 10: The strength knob on the randomised- r adapter at $r=0.1$, swept finely. Each scaled point below 1.00 pays for lower structure consistency with FID; over $r \in [0.05, 0.3]$ the cutoff dial keeps FID between 51.58 and 54.68 (Table 8). Scales 0.25, 0.5 and 0.75 were repeated with sampling seed 2027 on the same checkpoint.

r	specialist	one model, 3 seeds			
		mean	sd	difference	diff./sd
0.10	0.7049	0.6245	0.0059	+0.0803	+13.55
0.30	0.1214	0.2318	0.0106	-0.1104	-10.42
0.50	0.1211	0.1304	0.0093	-0.0093	-1.00
0.70	0.1287	0.1261	0.0186	+0.0026	+0.14

Table 11: Per-cutoff specialists (one training run each, seed 2026) against the single randomised- r model averaged over training seeds 2026–2028, each at its own cutoff. Difference is specialist minus the three-seed mean; the last column divides it by the three-seed standard deviation, as in Sec. 4.4. A positive difference favours the specialist. The specialist wins at $r=0.1$, the single model wins at $r=0.3$, and neither separates at $r=0.5$ or 0.7 ; what the single model gives that no specialist can is the rest of the curve.

r	out of domain			FFHQ test		
	vs. input	vs. filtered	FID	vs. input	vs. filtered	FID
0.05	0.4506	0.3139	128.40	0.5981	0.3193	61.07
0.10	0.3106	0.1830	148.97	0.5057	0.1796	65.71
0.20	0.1944	0.0873	154.24	0.4407	0.0799	68.78
0.30	0.1384	0.0303	160.82	0.4142	0.0334	70.75

Table 12: 500 general-scene photographs against 500 FFHQ test images, run with the same script and the prompt “a high-quality photo” (Sec. 4.6). “Vs. input” correlates the grayscale output with the grayscale input photograph, “vs. filtered” with the filtered condition the model was given; neither filters the output, so these are not the structure-consistency values of Table 1. Each FID is against that run’s own input photographs. The control falls in both; the image quality does not transfer.

r	40 face sketches			FFHQ test photos	
	vs. original	$\pm$ SEM	vs. filtered	vs. original	vs. filtered
0.00	0.5486	0.0273	0.5486	—	—
0.05	0.4826	0.0295	0.2552	0.5981	0.3193
0.10	0.3390	0.0356	0.1399	0.5057	0.1796
0.20	0.2746	0.0360	0.0590	0.4407	0.0799
0.30	0.2271	0.0364	0.0142	0.4142	0.0334
<i>beyond the range we claim</i>					
0.50	0.1379	0.0419	0.0013	—	—
0.70	0.1232	0.0419	0.0005	—	—
1.00	0.0407	0.0326	0.0000	—	—

Table 13: The dial applied to 40 hand-drawn face sketches (main model, sampling seed 2026, prompt “a high-quality photo of a face”). “Vs. original” correlates the grayscale output with the unfiltered drawing; “vs. filtered” with the drawing filtered at r that the model was given. Neither filters the output, so neither is the structure-consistency instrument of Table 1. The photograph columns come from the same script on FFHQ test images with the prompt “a high-quality photo” (Table 12). The $r=0$ row is an independent re-run of the zero-shot evaluation of Figure 6 (0.5484 there). No specialist was run on sketches.

Item	Setting
Base model	stable-diffusion-v1-5/stable-diffusion-v1-5, frozen
Condition encoder	seven 3×3 convolutions, SiLU, three with stride 2 and group normalisation, 32 to 128 channels, 4 output channels
Trainable params	6,611,469 (main); skip only 6,205,856; attention only 405,613
Data	FFHQ-512, 65,100 / 1,400 / 3,500, seeded shuffle (seed 2026); the validation split only monitors training on four images
Optimiser	AdamW, lr 10^{-4} , $\beta=(0.9, 0.999)$, weight decay 0.01, cosine, 500 warm-up steps
Batch / precision	32 / bf16, gradient clipping 1.0, 12 epochs, Min-SNR $\gamma=5$, one H200
EMA	decay 0.9999 every 10 steps; evaluation uses the raw weights
Dropout	text 0.1, structure condition 0.05
Training seeds	main model 2026, 2027, 2028; specialists, ablations and skip only 2026
Sampling	DDIM, 50 steps, CFG 7.5, seed 2026; Diversity seeds $2026+10^5+i$ for image i
Evaluation	first 500 of the 3,500 test images; clean-FID, clean mode, 512 px; LPIPS AlexNet against the reference; CLIP ViT-B/32 (OpenAI weights), $100\times$ cosine
Baselines	llyasviel/sd-controlnet-canny, llyasviel/sd-controlnet-hed, Ten-centARC/t2iadapter_canny_sd15v2; Canny 100/200 for all; conditioning scale 1.0; default PNDM sampler, 50 steps, CFG 7.5, seed 2026
Software	Python 3.12.3, PyTorch 2.10 (NGC 25.11), diffusers 0.32.2, clean-fid 0.1.35, lpips 0.1.4, open_clip 2.30.0

Table 14: Settings behind every table, read from the run manifests and evaluation scripts.